

THISUL HERATH

Artificial Intelligence Undergraduate

📍 607/20, The City, Jayanthi Rd,
Athurugiriya
☎ 0726206541
✉ thisulh@gmail.com
👤 <https://github.com/ThisulHerath>
🌐 <https://www.linkedin.com/in/thisul/>

IT Undergraduate specializing in Artificial Intelligence with hands-on experience in full-stack development and machine learning. Proficient in building web applications via the MERN stack and mobile solutions using React Native. Skilled in Python for AI/ML (model training & preprocessing) and Java Spring Boot for backend architecture. Proven ability to integrate intelligent components into scalable software. Seeking an internship to leverage technical expertise in Software Engineering and AI/ML.

PROJECTS

► Web Based Retail Inventory and Sales Forecasting System with Intelligent Analytics

February 2026 – April 2026

Technologies and Tools: React, Node.js, Express.js, Supabase (PostgreSQL), Python, Scikit-learn, JWT

- Engineered a full-stack retail platform using the PERN stack and Supabase Row Level Security (RLS) to provide real-time inventory visibility and secure multi-role access control.
- Integrated an AI forecasting server built with Python and Scikit-learn to generate sales predictions and trend analytics, enabling data-driven stock management.
- Developed a comprehensive RESTful API for POS operations, customer loyalty tracking, and automated invoice generation, ensuring high operational speed and data integrity.

► Wildlife Safari Trip Management System

July 2025 – September 2025

Technologies and Tools: Java (Spring Boot), MySQL, Hibernate, Maven, JavaScript, Bootstrap.

- Architected a full-stack web application to digitize safari operations, implementing MVC architecture to manage real-time bookings, vehicle allocations, and guide scheduling
- Developed a secure multi-role authentication system for Admins, Customers, and Staff, ensuring data integrity and role-based access control across finance and operations modules.
- Designed and optimized a MySQL database schema via Hibernate to handle complex entity relationships, reducing manual processing errors and improving resource tracking efficiency.

► Toxic-comment-detection

July 2025 – September 2025

Technology and Tools: Python, TensorFlow, Pandas, NLTK, TF-IDF, PCA, SVM, Scikit-learn, Matplotlib

- Developed an end-to-end NLP pipeline to classify toxic language in large-scale datasets, implementing custom text cleaning, tokenization, and stopword removal.
- Optimized model performance by applying TF-IDF vectorization and Principal Component Analysis (PCA) for dimensionality reduction, significantly improving training speed.
- Addressed class imbalance using RandomOverSampler and utilized GridSearchCV for hyperparameter tuning of SVM/LinearSVC models to maximize F1-score.

► Online Bookstore Management System

February 2025 – April 2025

Technologies and Tools: Java (Spring Boot), IntelliJ IDEA, HTML, CSS, JavaScript

- Built a web-based bookstore application using Java and OOP principles to manage books, users, and transactions.
- Implemented CRUD operations, file handling, and custom data structures (Linked List) for efficient data management.

► Home Automation System

July 2024 – September 2024

Technology and Tools: Arduino, Embedded C, Bluetooth Module (HC-05), Mobile Application, Arduino IDE

- Developed an Arduino-based home automation system to control lights and electrical devices using a mobile application via Bluetooth.
- Implemented real-time control features to improve convenience, energy efficiency, and smart home functionality.

EDUCATION

- BSc (Hons) in Information Technology Specializing in Artificial Intelligence | SLIIT | 2024 – 2028
- G.C.E. A/L | Royal College Colombo 07 | 2021 – 2024
- G.C.E. O/L | Royal College Colombo 07 | 2019 – 2021

CERTIFICATIONS

Certificate in AI/ML Engineer - SLIIT

- Stage 1 - <https://code.sliit.org/certificates/nddgg8tgtd>
- Stage 2 - <https://code.sliit.org/certificates/cbxmlypplk>

SKILLS

Technical Skills

- Languages: Java | Python | SQL | JavaScript | HTML | CSS | R
- Frameworks & Libraries: React | React Native | Node.js | Expressjs | Java Spring Boot | Pandas | TensorFlow
- AI & Machine Learning: Machine Learning | AI | NLP | Scikit-learn | Hyperparameter Tuning
- Tools & Databases: MongoDB | Supabase | PostgreSQL | MySQL | Postman | Git | GitHub | OOP

Soft Skills

- Problem Solving | Teamwork | Adaptability | Time Management | Communication | Continuous Learning

EXTRA-CURRICULAR ACTIVITY

Athletic Achievements & Colors

- Western Province Colors: Awarded for Badminton (2022, 2023) in recognition of provincial-level excellence
- School Colors (Royal College): Triple recipient of School Colors for Badminton (2019, 2022, 2023).
- House Captain (Badminton): Appointed as House Captain in 2023, overseeing team strategy and player development
- Vice Captain (U15 Team): Served as Vice Captain of the Royal College Under-15 Badminton team (2018).

Leadership & Service

- Junior Prefect: Appointed as a Junior Prefect at Royal College (2018), responsible for maintaining student discipline and assisting in school-wide event management.

REFERENCES

Nadith Kudagama

Chief Operating Officer/Director

Programus LTD

nadith@programus.co.uk | +44 (0) 748 087 0527

Sanjee Balasuriya

Chief Information Security Officer

Central Bank of Sri Lanka

sanjee@cbsl.lk | +94 77 577 0982